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Late Cretaceous palaeomagnetic results from Southwest Japan: new insights for early Cenozoic clockwise rotation

Koji Uno

Department of Environmental Sciences, Faculty of Integrated Human Studies, Kyoto University, Sakyo-ku, Kyoto 606-8501, *Japan.*
E-mail: unokoji@gaia.h.kyoto-u.ac.jp

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SUMMARY

Palaeomagnetic samples in the Upper Cretaceous Arima Group have been collected from two areas in the middle of Southwest Japan. Tuffaceous sediments were sampled at eleven sites in the Arima area and welded tuffs were sampled at four sites in the Ikuno area. Thermal demagnetization isolated characteristic remanent magnetization components with unblocking temperatures of 500–580°C from all the sites. After tilt correction, the characteristic directions provide Late Cretaceous mean directions for the Arima area ($D = 46.6°$, $I = 53.7°$, $\alpha_{95} = 3.7°$) and the Ikuno area ($D = 78.4°$, $I = 44.7°$, $\alpha_{95} = 9.5°$). I interpret these data to show that the palaeomagnetic pole from the Arima area has been rotated counter-clockwise relative to those from the central part of Southwest Japan, while the pole from the Ikuno area is in good agreement with palaeomagnetic poles from the central part of Southwest Japan. The pole for the Ikuno area together with previously reported ones give a new Late Cretaceous pole representative of the central part of Southwest Japan. This pole is rotated $64.6° \pm 15.5°$ clockwise with respect to the coeval one for the North China Block. In my model, the tectonic history of Southwest Japan includes Early Tertiary clockwise rotation of about 23° and Middle Miocene clockwise rotation of about 42°, resulting in a total rotation of Southwest Japan by 65°. The Early Tertiary clockwise rotation occurred in an area incorporating the Korean Peninsula and Southwest Japan, and the Middle Miocene clockwise rotation was associated with the opening of the Japan Sea.

Key words: Cretaceous, palaeomagnetism, Southwest Japan.

1 INTRODUCTION

The Japanese Islands are located in the easternmost part of the Asian continent facing oceanic plates to the east (Fig. 1a). The Japanese Islands have been recognized as continental slivers rifted from the Asian continent, associated with the opening of the Japan Sea (Otofuji *et al.* 1985). As a consequence of the rifting, Southwest and Northeast Japan experienced clockwise and counter-clockwise rotations respectively (Otofuji & Matsuda 1987; Otofuji *et al.* 1991, 1994; Jolivet *et al.* 1994). The amount of rotation in the eastern and western parts of Southwest Japan (Fig. 1a) is reduced relative to that in the central part of Southwest Japan (Faure & Lalevée 1987; Itoh 1988; Ishikawa 1997).

The clockwise rotation of Southwest Japan in response to the opening of the Japan Sea occurred during the Middle Miocene. Otofuji & Matsuda (1987) suggested that more than 80 per cent of the net rotation occurred later than 16.1 Ma and ceased by 14.2 Ma. At about 15 Ma Southwest Japan was at the peak of rotational motion (Hayashida 1986; Otofuji *et al.* 1991). This timing of rotation from palaeomagnetism is concordant with those from geochronological data, marine magnetic anomaly patterns and ODP Leg 127 drilling results (Seama & Isezaki 1990; Kaneoka *et al.* 1990; Shipboard Scientific Party Leg 127 and Leg 128 1990).

Early Miocene palaeomagnetic directions from Southwest Japan show consistent declinations of about 40–50° (Hayashida & Ito 1984; Hayashida 1986; Hayashida *et al.* 1991; Otofuji *et al.* 1991; Kojima *et al.* 2001). On the other hand, Late Cretaceous palaeomagnetic directions from Southwest Japan have larger declination values of about 70° (Otofuji & Matsuda 1987). There has been an issue that the Middle Miocene tectonic event does not account for the discrepancy in declinations between Late Cretaceous and Early Miocene times. Two factors are relevant to this discrepancy. One is that the Cretaceous palaeomagnetic pole for the Asian continent (North China Block) (Gilder & Courtillot 1997) gives an easterly-deflected expected declination at the reference point of Southwest Japan. Another is the clockwise rotation of the Korean Peninsula with respect to the North China Block in Early Tertiary time (Ma *et al.* 1993; Uchimura *et al.* 1996; Uno 2000). Because Southwest Japan is considered to have been flanked on the eastern side of the Korean Peninsula at this time (Otofuji & Matsuda 1983), this clockwise rotation may be superimposed on Cretaceous palaeomagnetic declinations from Southwest Japan. If the amount of rotation

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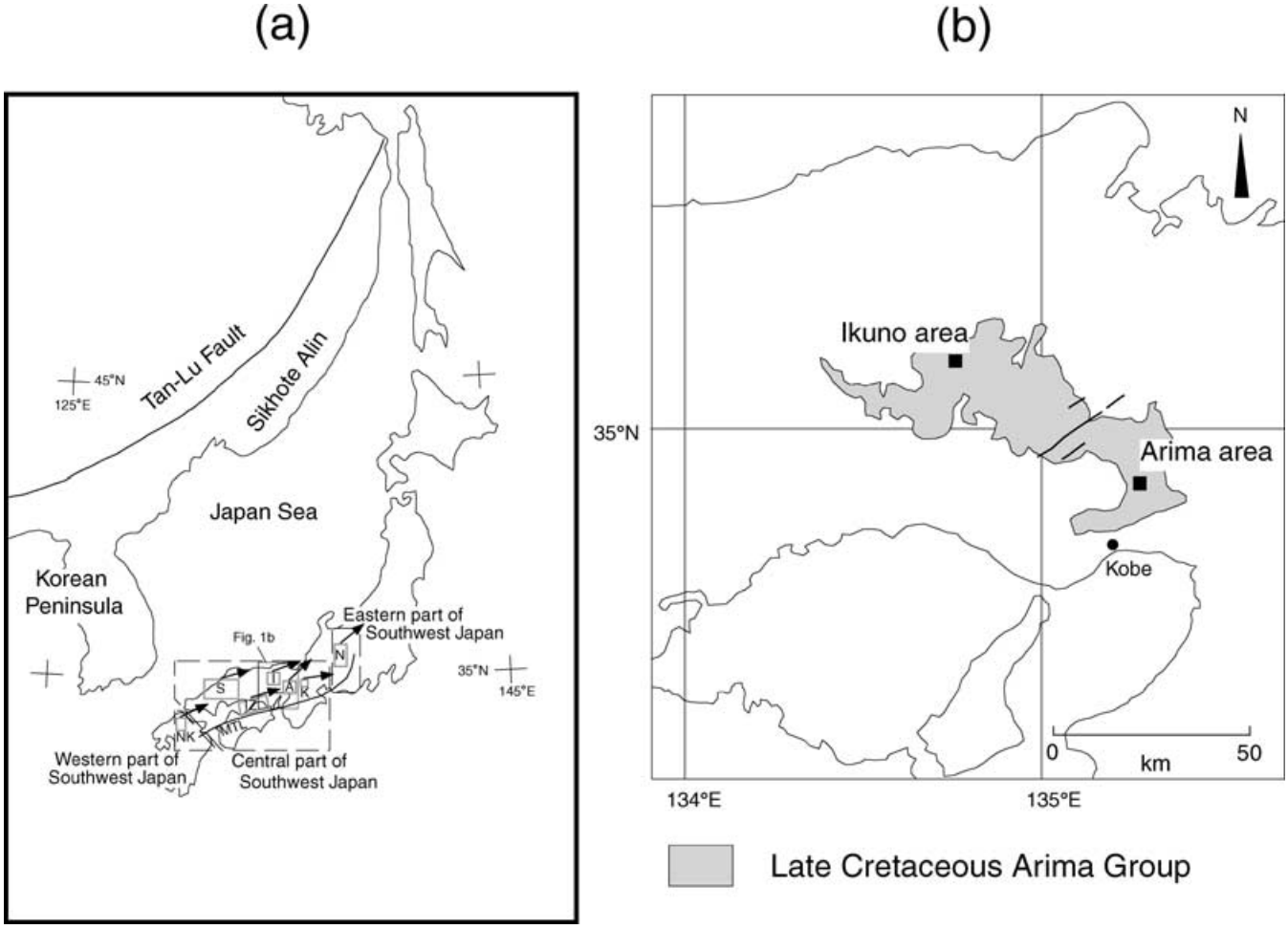


Figure 1. (a) Sketch map of the Japan arc and adjacent area. N: Nohi area; K: Koto area; A: Arima area (study area); I: Ikuno area (study area); IZ: Izumi area; S: San'in area; NK: North Kyushu area. Arrows are Cretaceous palaeomagnetic directions from Southwest Japan. MTL denotes the Median Tectonic Line. (b) Map showing the distribution area of the Arima Group and study areas.

between the North China Block and Southwest Japan recorded in Late Cretaceous data is larger than that recorded in Early Miocene data, Southwest Japan as well as the Korean Peninsula probably underwent tectonic rotation with respect to the North China Block in Early Tertiary times.

At present, a Late Cretaceous pole from one locality (the San'in area, Fig. 1a) (Otofuji & Matsuda 1987) is regarded as the representative one for Southwest Japan. This pioneering palaeomagnetic study of Cretaceous rocks in Southwest Japan (Otofuji & Matsuda 1987) was not followed by any palaeomagnetic studies with a view to establishing the representative pole for Southwest Japan. More sampling is required to improve the Late Cretaceous pole position for Southwest Japan with which to make a comparison with those from the North China Block and the Korean Peninsula.

This paper focuses on the Arima and Ikuno areas in the middle of Southwest Japan because this region lacks palaeomagnetic data (Fig. 1a). The stratigraphic sampling target for this palaeomagnetic study is the Late Cretaceous Arima Group, which is distributed in both Arima and Ikuno areas. I present new palaeomagnetic data from the two areas and a Late Cretaceous palaeomagnetic pole for the central part of Southwest Japan. This is the first report dealing with Japanese Mesozoic data in the light of tectonics of the East Asian continent. The Arima and Ikuno areas are 60 km apart from each other.

2 GEOLOGICAL SETTING AND SAMPLING

The Late Cretaceous Arima Group unconformably overlies the Jurassic–Early Cretaceous Tamba Group and the Early Cretaceous Sasayama Group, and is unconformably overlain by the Eocene–Oligocene Kobe Group. The Arima Group is well exposed in the Arima (34.8°N, 135.3°E) and Ikuno (35.2°N, 134.7°E) areas to the north of Kobe (Fig. 1b). It is made up mainly of rhyolitic lavas and pyroclastic rocks, and is divided into three parts (lower, middle and upper formations) (Kasama & Yoshida 1976; Nakazawa *et al.* 1987). The upper formation of the Arima Group is K–Ar dated at 72.1 ± 3.6 Ma (Yoshida & Kawada 1987). Tilting of the Arima Group occurred prior to the Eocene.

In the Arima area, all the three formations are composed mainly of welded tuffs. Welded tuffs in the middle and upper formations of the Arima area are rich in lithic fragments, whereas those in the lower formation rarely contain lithic fragments (Kasama & Yoshida 1976; Nakazawa *et al.* 1987). Eutaxitic structures in welded tuffs are not observable in this area. The three formations are demarcated by two tuffaceous lake deposits of about 200 m in thickness: the Sogawa sedimentary rocks demarcate the lower and middle formations, and the Gen'noike sedimentary rocks demarcate the middle and upper formations. They dip 12–22° northward

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in the study area, with roughly consistent dip directions of 310–360°.

In the Ikuno area, the lower and middle formations are composed mainly of volcanic breccias and lavas (Nakazawa *et al.* 1987). Welded tuffs are found in the upper formation. Eutaxitic structures are observed at some outcrops in this formation. The tilt of eutaxitic structure is gentle for all the sites (<20°), with dip directions of 110° and 310–330°.

Tuffaceous mudstones and sandstones in the Sogawa (sites AG1–6 and AG12) and Gen'noike (sites AG8–11) sedimentary rocks were collected at eleven sites in the Arima area, and welded tuffs in the upper formation (sites IG1–4) were collected at four sites in the Ikuno area (Fig. 1b). In each site, six to eight hand samples were distributed over distances ranging up to 10 m and were oriented by magnetic compass.

3 PALAEOMAGNETISM

In the laboratory, cores of 25 mm in diameter and 22 mm long were prepared from each hand sample. Natural remanent magnetizations (NRMs) were measured with a 2-G Superconducting Rock Magnetometer (SRM). All samples were subjected to progressive thermal demagnetization treatment with a Natsuhara TDS-1 thermal demagnetizer. Results for each sample were plotted on orthogonal vector diagrams (Zijderveld 1967) to evaluate the demagnetization behaviours (Fig. 2). Principal component analysis (Kirschvink 1980) was used to estimate directions of the magnetic components (Table 1).

Samples from the Arima area have initial NRM intensities ranging from 2×10^{-3} to 3×10^{-4} Am^{-1}. Thermal demagnetization revealed a stable remanent magnetization component following removal of a component at around 260°C (Figs 2a–e). The high-temperature component is well-defined and generally resolved by demagnetization steps above 260°C, where more than half of the initial NRM persists. For each sample, the high-temperature component with an unblocking temperature of 500°C is regarded as a characteristic remanent magnetization component, which is identified from all the sites.

The site-mean directions of the characteristic component cluster well at each site: the precision parameter (k-value) ranges from 74.0 to 461.4. These directions are of east declination and relatively steep inclination before tilt correction, and of northeast declination and intermediate positive inclination after tilt correction (Fig. 3). Mean directions of the 11 sites before and after tilt correction are $D = 69.6°$, $I = 57.8°$ with $\alpha_{95} = 2.9°$ and $D = 46.6°$, $I = 53.7°$ with $\alpha_{95} = 3.7°$, respectively. The timing of remanence acquisition could not be established by the fold test alone, because the bedding attitudes are essentially monoclinal. Although the precision parameter decreases from 253.8 to 151.1 after tilt correction, the decrease is not quite demonstrably significant for a negative fold test at the 95 per cent confidence level (McElhinny 1964).

The characteristic component of the Arima Group is not attributed to a secondary origin because of the following points. (1) The clear demarcation between the characteristic component and lower-temperature component in orthogonal vector diagram suggests little overlap of unblocking temperature spectra between these components. (2) The 11 site-mean directions before tilt correction are significantly removed from the present Earth's field direction ($D = 352.9°$, $I = 49.0°$) and axial dipole field direction ($D = 0.0°$, $I = 54.3°$). (3) A thermal perturbation in the Arima area is restricted to a small area in its southeastern periphery, as evidenced by distribution of biotite hornfels (Kasama & Yoshida 1976).

Samples from the Ikuno area have initial NRM intensities ranging from 5 Am^{-1} to 4×10^{-2} Am^{-1}. With the exception of a component removed in the first three or four steps of treatment (260–320°C), thermal demagnetization isolated a stable remanent magnetization component with an unblocking temperature of 580°C, suggesting that it resides in magnetite. The high-temperature component decays univectorially to the origin and is regarded as a characteristic remanent magnetization component, which is identified from all the sites.

The site-mean directions of the characteristic component have the precision parameter ranging from 32.9 to 155.0. The four site-mean directions before tilt correction, with a mean of $D = 264.3°$, $I = -44.2°$ with $\alpha_{95} = 16.4°$, show westerly declinations and variable inclinations between $-25°$ and $-56°$, while these directions after tilt correction, with a mean of $D = 258.4°$, $I = -44.7°$ with $\alpha_{95} = 9.5°$, have westerly declinations with intermediate negative inclinations (Fig. 4). Tilt correction increases the precision parameter of the mean direction from 32.6 to 93.9. The precision parameter after tilt correction is 2.9 times greater than before tilt correction. Although the fold test is not significant at the 95 per cent confidence level (McElhinny 1964), the improvement of the clustering suggests that the characteristic component is likely primary magnetization.

The tilt-corrected mean direction of the Ikuno area, presumably of primary origin, is compared with the mean directions of the Arima area before and after tilt correction. An inclination contrast between the two areas may be calculated to test whether or not the Arima area has been remagnetized. With respect to the tilt-corrected mean direction of the Ikuno area, the *in situ* mean direction of the Arima area shows an inclination difference of $13.6° \pm 9.9°$, while the tilt-corrected one shows an inclination difference of $9.0° \pm 10.2°$. Because the difference in inclination becomes insignificant after tilt correction, I interpreted this observation to show that the characteristic component of the Arima area is not of secondary magnetization.

An alternative explanation of the characteristic direction of the Arima area is remagnetization. The *in situ* direction of the Arima area is closer to the tilt-corrected direction of the San'in area (Otofuji & Matsuda 1987) than the tilt-corrected Arima direction: the *in situ* Arima direction differs by $14.2° \pm 10.4°$ from the coeval San'in direction, while the tilt-corrected Arima direction differs by $20.5° \pm 10.7°$ from the San'in direction. It may be possible that the Arima area suffered complete remagnetization after tilting (Late Cretaceous–Eocene) and prior to the opening of the Japan Sea (~15 Ma).

4 DISCUSSION

Palaeomagnetic poles for the Arima and Ikuno areas are calculated on the basis of the tilt-corrected characteristic directions from each area. They are located at 52.2°N, 211.8°E, $A_{95} = 3.8°$ and 24.1°N, 210.0°E, $A_{95} = 8.5°$, respectively (Table 2). Previously reported Late Cretaceous poles for Southwest Japan are available from the Nohi area (eastern part of Southwest Japan, Otofuji *et al.* 1999), Koto area (central part of Southwest Japan, Fukuma & Torii 1990), Izumi area (central part of Southwest Japan, Kodama 1990), San'in area (central part of Southwest Japan, Otofuji & Matsuda 1987) and North Kyushu area (western part of Southwest Japan, Ishikawa 1997) (Fig. 1a). A Cretaceous reference pole for the North China Block lies at 78.6°N, 202.6°E with $A_{95} = 6.2°$ (Gilder & Courtillot 1997), and a Cretaceous pole for the Korean Peninsula falls at 61.2°N, 199.5°E with $A_{95} = 6.6°$ (Zhao *et al.* 1999).

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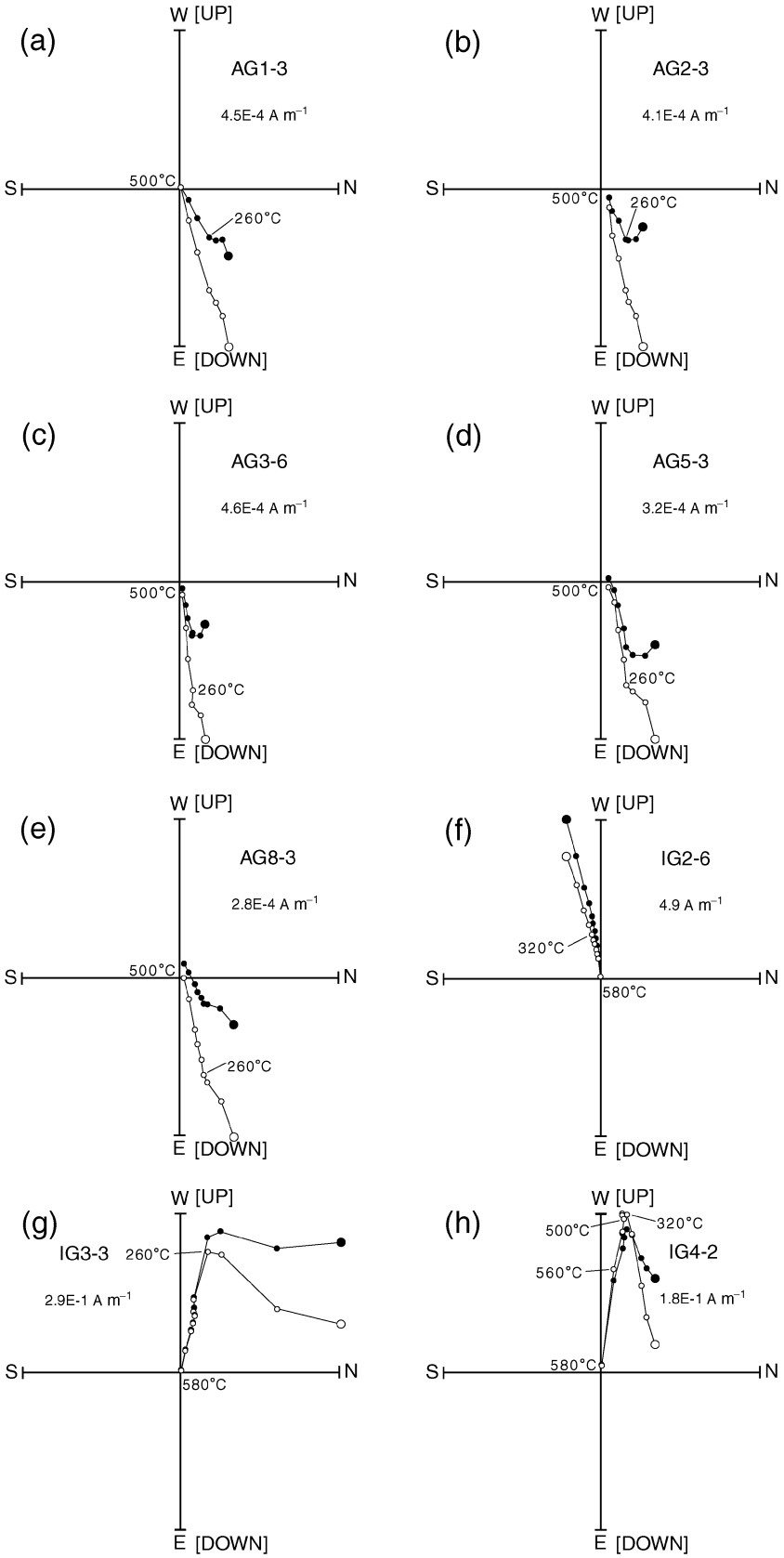


Figure 2. Orthogonal vector plots of thermal demagnetization of the Arima Group in the Arima area (a–e) and the Ikuno area (f–h) before tilt correction. Solid and open symbols represent projection on horizontal and vertical planes, respectively.

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Table 1. Palaeomagnetic results of the Arima Group in the Arima and Ikuno areas.

Site	Lithology	N	*In situ*		Tilt corrected		k	α_{95} (°)	Strike Dip DD		
			D (°)	I (°)	D (°)	I (°)			(°)	(°)	(°)
Arima area											
AG1	TM	6	72.7	58.5	44.3	50.9	114.0	6.3	259	21	349
AG2	TM	6	85.3	61.1	49.2	56.0	133.0	5.8	260	22	350
AG3	TM	6	78.4	63.1	47.5	54.6	179.4	5.0	267	20	357
AG4	TM	6	65.4	52.4	46.9	44.4	87.4	7.2	262	18	352
AG5	TM	6	68.9	56.9	47.5	48.6	167.0	5.2	265	18	355
AG6	TM	5	69.9	51.2	51.0	43.1	74.0	9.0	265	19	355
AG8	TM	6	61.8	61.7	38.4	57.8	138.3	5.7	246	14	336
AG9	TS	6	69.9	56.8	51.1	58.2	210.5	4.2	234	12	324
AG10	TS	4	69.0	57.3	50.5	56.6	229.7	6.1	243	12	333
AG11	TS	4	63.1	56.2	40.1	56.9	461.4	4.3	229	15	319
AG12	TM	6	64.4	58.8	43.7	62.6	138.0	5.7	216	12	306
Mean		11	69.6	57.8			253.8	2.9			
					46.6	53.7	151.1	3.7			
Ikuno area											
IG1	WT	7	250.5	−55.5	260.9	−39.0	69.9	7.3	16	19	106
IG2	WT	7	264.4	−47.7	250.9	−51.3	51.2	8.5	241	12	331
IG3	WT	4	267.3	−25.4	259.8	−37.1	32.9	16.3	220	17	310
IG4	WT	7	271.4	−47.0	260.1	−51.3	155.0	4.9	241	10	331
Mean		4	264.3	−44.2			32.6	16.4			
					258.4	−44.7	93.9	9.5			

TM: tuffaceous mudstones; TS: tuffaceous sandstones; WT: welded tuffs; N: number of the samples in calculation for the means; D and I: site-mean declination and inclination; k: the precision parameter; α_{95}: radius of the cone of 95 per cent confidence; DD: dip direction.

The pole for the Arima area is plotted to a lower latitudinal position than the reference pole for the North China Block (Fig. 5). The angular distance between the poles is $26.6^\circ \pm 7.3^\circ$. The difference between the two poles represents a clockwise rotation of the Arima pole through $33.3^\circ \pm 11.2^\circ$ with respect to that for the North China Block at the reference point of Southwest Japan (35°N, 134°E). The pole for the Ikuno area differs by $54.6^\circ \pm 10.5^\circ$ from that for the North China Block. The Ikuno pole is rotated clockwise through $64.5^\circ \pm 16.4^\circ$ with respect to that for the North China Block.

There appears to be two groups of Late Cretaceous poles from Southwest Japan (Fig. 5): those with smaller clockwise deflection of about 35° from the reference pole for the North China Block, and those with larger clockwise deflection of about 65°. On the one hand, the Arima pole well agrees with the pole for the Nohi area at 47.8°N, 211.3°E with $A_{95} = 9.0^\circ$. The Nohi pole is rotated clockwise through $38.1^\circ \pm 17.0^\circ$ with respect to that for the North China Block. This amount of rotation is consistent with that for the Arima area within confidence limits. On the other hand, the Ikuno pole is close to those for the Koto, Izumi and San'in areas (Table 2). The Koto, Izumi and San'in poles show clockwise rotations of $71.2^\circ \pm 16.3^\circ$, $65.9^\circ \pm 13.2^\circ$ and $60.0^\circ \pm 16.6^\circ$ with respect to that for the North China Block, respectively. The pole for

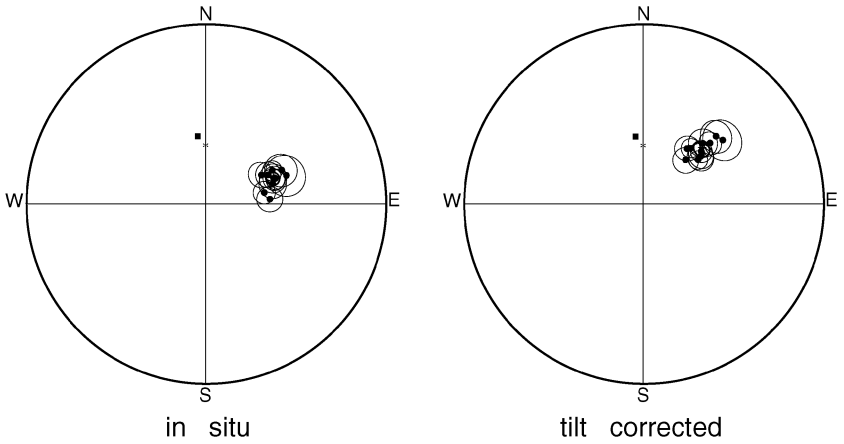


Figure 3. Equal area projections of the directions of the characteristic component (with associated 95 per cent confidence limits) from 11 sites of the Arima Group in the Arima area. The star and bold circle represent the mean direction and associated 95 per cent confidence limit, respectively. The solid square and asterisk represent the present Earth's field direction and axial dipole field direction, respectively.

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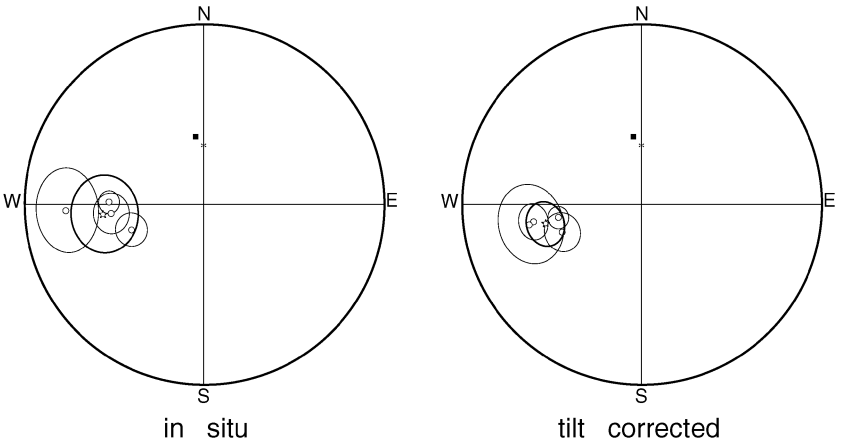


Figure 4. Equal area projections of the directions of the characteristic component (with associated 95 per cent confidence limits) from four sites of the Arima Group in the Ikuno area. The star and bold circle represent the mean direction and associated 95 per cent confidence limit, respectively. The solid square and asterisk represent the present Earth's field direction and axial dipole field direction, respectively.

the North Kyushu area (38.8°N, 197.6°E, $A_{95} = 28.9°$) shows an intermediate value of clockwise rotation through 52.3°; however, the fairly large uncertainty in pole position (28.9°) makes the comparison with other poles difficult.

The consistent results from palaeomagnetic poles from the central part of Southwest Japan suggest that the central part of Southwest Japan essentially have a single representative pole for Late Cretaceous time. The palaeomagnetic poles from the Koto, Ikuno, Izumi and San'in areas are averaged to yield a Late Cretaceous palaeomagnetic pole for the central part of Southwest Japan (25.8°N, 203.8°E with $A_{95} = 7.5°$). This mean pole covers a distance of about 350 km in the central part of Southwest Japan. Reduction in clockwise deflection of the pole for the Nohi area can be ascribed to counter-clockwise rotation of the eastern part of Southwest Japan with respect to the central part (Itoh 1988; Otofuji *et al.* 1999), as evidenced by curvature of the Median Tectonic Line (MTL, Fig. 1a), and that for the Arima area may be attributed to internal local rotation of this area with counter-clockwise sense, which is observed as a result of sinistral faulting in the northwestern part of the Arima area (Fig. 1b) (Yoshida & Kawada 1987; Yoshikawa 1993).

The Late Cretaceous palaeomagnetic pole for the central part of Southwest Japan is significantly removed by 52.8° ± 9.7° from the Cretaceous one for the North China Block (Gilder & Courtillot 1997). The difference in pole position between the two is ascribed to clockwise rotation of Southwest Japan through 64.6° ± 15.5° with respect to the North China Block, because no significant palaeolatitude contrast (9.0° ± 9.5°) is present between the observed and expected palaeolatitudes at the reference point of Southwest Japan. Clockwise rotation occurred after the Late Cretaceous.

The Cretaceous pole for the Korean Peninsula (Zhao *et al.* 1999) differs by 17.4° ± 9.1° from the coeval one for the North China Block, and is rotated clockwise through 22.6° ± 16.5° with respect to that for the North China Block. No significant palaeolatitude difference exists between the two regions (3.2° ± 8.3°). The Korean Peninsula underwent clockwise rotation with respect to the North China Block after the Cretaceous. The clockwise rotation of the Korean Peninsula probably occurred in Early Tertiary times (Ma *et al.* 1993; Uchimura *et al.* 1996; Uno 2000): this timing of rotation is constrained by the fact that the Korean Peninsula, as well as Sikhote Alin (Fig. 1a), was tectonically quiescent in terms of rotation during the Middle Miocene opening of the Japan Sea, with the exception of local tectonic rotation caused by fault motion in the southeastern tip of the Korean Peninsula (Otofuji *et al.* 1995; Lee *et al.* 1999).

Table 2. Cretaceous palaeomagnetic poles for Southwest Japan and East Asia.

Locality (Area)	*N* (*n*)	Pole		A_{95} (°)	Rotation (°)	Reference
		Lat. (°N)	Long. (°E)			
Southwest Japan						
(Nohi area)	(14)	47.8	211.3	9.0	38.1 ± 17.0	Otofuji *et al.* (1999)
(Koto area*)	(33)	25.6	194.1	7.1	71.2 ± 16.3	Fukuma & Torii (1990)
(Arima area)	(11)	52.2	211.8	3.8	33.3 ± 11.2	This study
(Ikuno area*)	(4)	24.1	208.0	8.5	64.5 ± 16.4	This study
(Izumi area*)	(70)	25.0	203.0	6.0	65.9 ± 13.2	Kodama (1990)
(San'in area*)	(14)	28.0	210.0	9.8	60.0 ± 16.6	Otofuji & Matsuda (1987)
(North Kyushu area)	(4)	38.8	197.6	28.9	52.3 ± 53.7	Ishikawa (1997)
Mean	4*	25.8	203.8	7.5	64.6 ± 15.5	This study
Korean Peninsula	9	61.2	199.5	6.6	22.6 ± 16.5	Zhao *et al.* (1999)
North China Block	5	78.6	202.6	6.2	—	Gilder & Courtillot (1997)

N (*n*): the number of the studies (sites); Lat.: north latitude; Long.: east longitude; A_{95}: radius of the circle of the 95 per cent confidence.

Rotation is evaluated by comparison between the observed and expected declinations at the reference point of Southwest Japan.

4* = Fisher average of Koto, Ikuno, Izumi and San'in poles.

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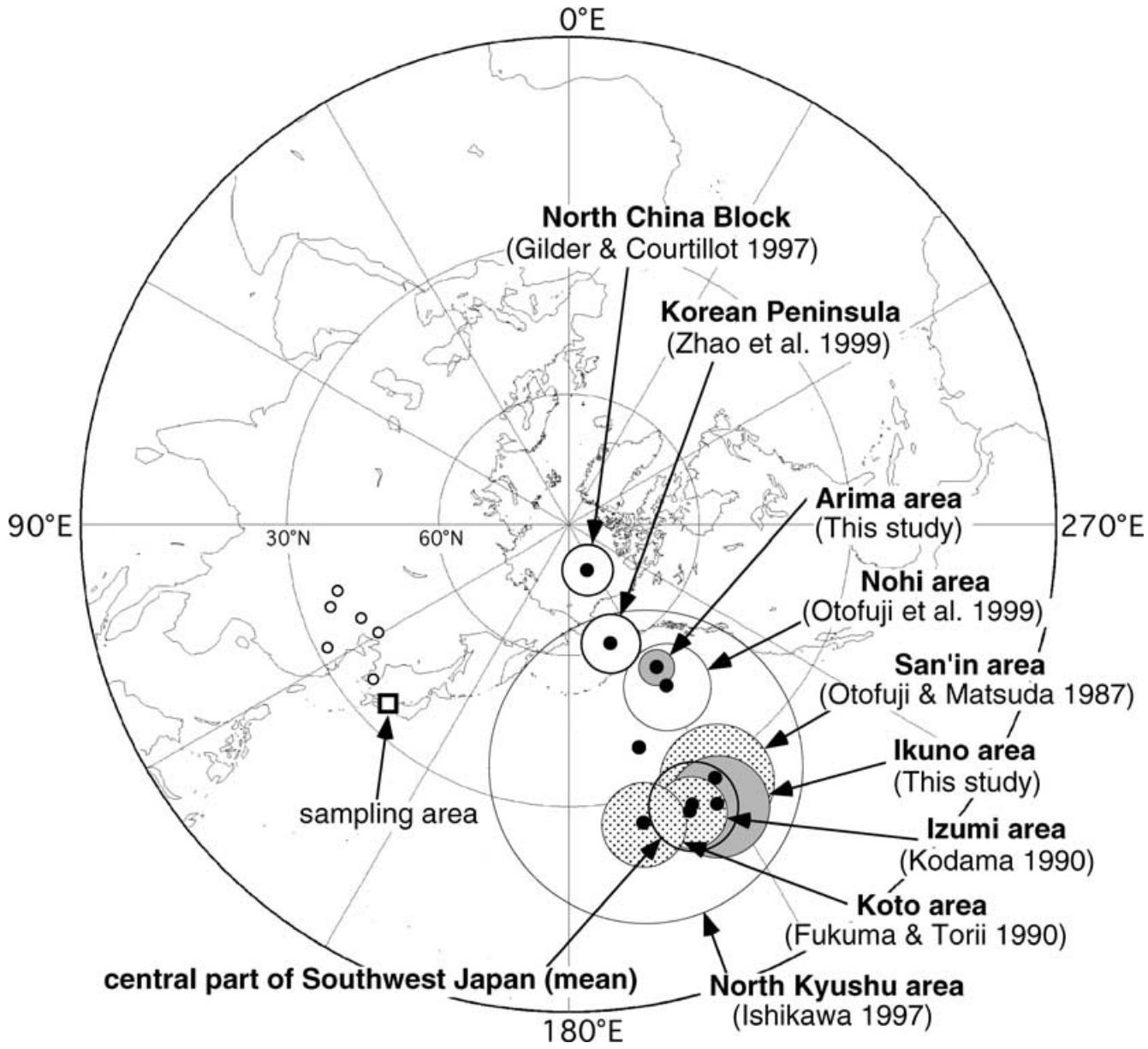


Figure 5. Cretaceous palaeomagnetic poles (with associated 95 per cent confidence limits) for Southwest Japan, North China Block and Korean Peninsula. The mean pole of Southwest Japan is calculated on the basis of four poles from the central part of Southwest Japan (Koto, Ikuno, Izumi and San'in areas). The open square and circles denote the reference point of Southwest Japan and the sites of reference palaeomagnetic studies, respectively. Please refer to Table 2 for numerical data.

Early Tertiary clockwise rotation of about 23° observed in data from the Korean Peninsula is also recorded in Late Cretaceous data from the central part of Southwest Japan. Rotation of an area incorporating the Korean Peninsula and Southwest Japan is anticipated for the reconstructions between the Korean Peninsula and Southwest Japan in Early Tertiary times: many authors claimed that Southwest Japan was flanked on the eastern side of the Korean Peninsula in this period (Otofuji & Matsuda 1983; Hayashida *et al.* 1991; Smith & Yamauchi 1994; Lee *et al.* 1999). If Southwest Japan, together with the Korean Peninsula, was subjected to Early Tertiary clockwise rotation by about 23°, a clockwise rotation of about 42° is expected to have occurred during the Middle Miocene opening of the Japan Sea because Southwest Japan records a total clockwise rotation of 65° (Table 2). According to Otofuji (1996), the amount of clockwise rotation associated with the Middle Miocene opening of the Japan Sea is determined to be 42°. There is good agreement between the expected and observed values of rotation. It is therefore concluded that Southwest Japan underwent Early Tertiary clockwise rotation by 23° and Middle Miocene clockwise rotation by 42°, resulting in a total rotation of Southwest Japan by 65°.

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